

Linear Discriminant Analysis (LDA)

1 Setup

Let $f_h(\mathbf{x})$ be the class-conditional density of $\mathbf{x}$ in class $G = h$ ($\mathbf{x} \in \mathbb{R}^p$).

Let π_h be the prior probability of class h , s.t. $\sum_{h=1}^K \pi_h = 1$.

By Bayes' theorem,

$$\mathbb{P}(G = h \mid \mathbf{X} = \mathbf{x}) = \frac{f_h(\mathbf{x}) \pi_h}{\sum_{\ell=1}^K f_\ell(\mathbf{x}) \pi_\ell}.$$

Suppose we model each class density as multivariate Gaussian with mean $\boldsymbol{\mu}_h$ and common covariance matrix Σ :

$$f_h(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_h)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_h)\right).$$

When comparing two classes, it is sufficient to look at the log-ratio:

$$\begin{aligned} \log\left(\frac{\mathbb{P}(G = h \mid \mathbf{X} = \mathbf{x})}{\mathbb{P}(G = \ell \mid \mathbf{X} = \mathbf{x})}\right) &= \log\left(\frac{f_h(\mathbf{x}) \pi_h}{f_\ell(\mathbf{x}) \pi_\ell}\right) \\ &= \log\left(\frac{\pi_h}{\pi_\ell}\right) - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_h)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_h) + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_\ell)' \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}_\ell) \\ &= \log\left(\frac{\pi_h}{\pi_\ell}\right) - \frac{1}{2} [-\mathbf{x}' \Sigma^{-1} \boldsymbol{\mu}_h - \boldsymbol{\mu}_h' \Sigma^{-1} \mathbf{x} + \boldsymbol{\mu}_h' \Sigma^{-1} \boldsymbol{\mu}_h] + \frac{1}{2} [-\mathbf{x}' \Sigma^{-1} \boldsymbol{\mu}_\ell - \boldsymbol{\mu}_\ell' \Sigma^{-1} \mathbf{x} + \boldsymbol{\mu}_\ell' \Sigma^{-1} \boldsymbol{\mu}_\ell] \\ &= \log\left(\frac{\pi_h}{\pi_\ell}\right) - \frac{1}{2} [\boldsymbol{\mu}_h' \Sigma^{-1} \boldsymbol{\mu}_h - \boldsymbol{\mu}_\ell' \Sigma^{-1} \boldsymbol{\mu}_\ell] + \mathbf{x}' \Sigma^{-1} (\boldsymbol{\mu}_h - \boldsymbol{\mu}_\ell) \\ &= \log\left(\frac{\pi_h}{\pi_\ell}\right) - \frac{1}{2} (\boldsymbol{\mu}_h + \boldsymbol{\mu}_\ell)' \Sigma^{-1} (\boldsymbol{\mu}_h - \boldsymbol{\mu}_\ell) + \mathbf{x}' \Sigma^{-1} (\boldsymbol{\mu}_h - \boldsymbol{\mu}_\ell), \end{aligned}$$

an equation *linear* in $\mathbf{x}$.

In practice we do not know $\pi_h, \boldsymbol{\mu}_h, \Sigma$, and estimate them using the training data.

Let N_h be the number of class- h observations in the training data, then

$$\hat{\pi}_h = N_h/N \quad \text{where } N = \sum_{h=1}^K N_h,$$

$$\hat{\boldsymbol{\mu}}_h = \sum_{i \in h} \mathbf{x}_i / N_h,$$

$$\hat{\Sigma} = \sum_{h=1}^K \sum_{i \in h} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h)' / (N - K).$$

2 LDA with two classes

With two classes, there is a simple correspondence between LDA and classification by linear regression. The LDA rule classifies to class 2 if

$$\log\left(\frac{N_2/N}{N_1/N}\right) - \frac{1}{2}\left(\sum_{i \in 2} \mathbf{x}_i/N_2 + \sum_{i \in 1} \mathbf{x}_i/N_1\right) \hat{\Sigma}^{-1}(\hat{\boldsymbol{\mu}}_2 - \hat{\boldsymbol{\mu}}_1) + \mathbf{x}' \hat{\Sigma}^{-1}(\hat{\boldsymbol{\mu}}_2 - \hat{\boldsymbol{\mu}}_1) > 0,$$

i.e.

$$\mathbf{x}' \hat{\Sigma}^{-1}(\hat{\boldsymbol{\mu}}_1 - \hat{\boldsymbol{\mu}}_2) > \frac{1}{2}(\hat{\boldsymbol{\mu}}_2 + \hat{\boldsymbol{\mu}}_1)' \hat{\Sigma}^{-1}(\hat{\boldsymbol{\mu}}_2 - \hat{\boldsymbol{\mu}}_1) - \log(N_2/N_1).$$

3 The Least Squares criterion

Now consider the Least Squares criterion:

$$L(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^N (y_i - \beta_0 - \mathbf{x}'_i \boldsymbol{\beta})^2, \quad \text{where } N = N_1 + N_2.$$

Let the targets be coded as $-N/N_1$ and N/N_2 , so that

$$\bar{y} = \frac{1}{N} \left[N_1(-N/N_1) + N_2(N/N_2) \right] = 0, \quad \text{hence:}$$

$$\begin{aligned} \partial_{\beta_0} L(\hat{\beta}_0, \hat{\boldsymbol{\beta}}) = 0 &\iff -2 \sum_{i=1}^N (y_i - \hat{\beta}_0 - \mathbf{x}'_i \hat{\boldsymbol{\beta}}) = 0 \\ &\iff N\bar{y} - N\hat{\beta}_0 - \frac{1}{N} N \sum_{i=1}^N \mathbf{x}'_i \hat{\boldsymbol{\beta}} = 0 \\ &\iff \hat{\beta}_0 = -\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \hat{\boldsymbol{\beta}} = -\bar{\mathbf{x}}' \hat{\boldsymbol{\beta}}. \end{aligned}$$

Then the residual for observation i is

$$\hat{\varepsilon}_i = y_i + \bar{\mathbf{x}}' \hat{\boldsymbol{\beta}} - \mathbf{x}'_i \hat{\boldsymbol{\beta}} = y_i - (\mathbf{x}_i - \bar{\mathbf{x}})' \hat{\boldsymbol{\beta}} = y_i - \mathbf{z}'_i \hat{\boldsymbol{\beta}}, \quad \text{where } \mathbf{z}_i = \mathbf{x}_i - \bar{\mathbf{x}}.$$

Then, the criterion becomes $\sum_{i=1}^N (y_i - \mathbf{z}'_i \hat{\boldsymbol{\beta}})^2$; differentiating,

$$-2 \sum_{i=1}^N \mathbf{z}_i (y_i - \mathbf{z}'_i \hat{\boldsymbol{\beta}}) = 0 \iff \left(\sum_{i=1}^N \mathbf{z}_i \mathbf{z}'_i \right) \hat{\boldsymbol{\beta}} = \sum_{i=1}^N \mathbf{z}_i y_i.$$

Simplifying the right-hand side

Where

$$\begin{aligned} \sum_{i=1}^N \mathbf{z}_i y_i &= \sum_{i \in 1} (\mathbf{x}_i - \bar{\mathbf{x}})(-N/N_1) + \sum_{i \in 2} (\mathbf{x}_i - \bar{\mathbf{x}})(N/N_2) \\ &= \left(-\frac{N}{N_1}\right) N_1(\hat{\boldsymbol{\mu}}_1 - \bar{\mathbf{x}}) + \frac{N}{N_2} N_2(\hat{\boldsymbol{\mu}}_2 - \bar{\mathbf{x}}) \\ &= -N(\hat{\boldsymbol{\mu}}_1 - \bar{\mathbf{x}}) + N(\hat{\boldsymbol{\mu}}_2 - \bar{\mathbf{x}}) = N(\hat{\boldsymbol{\mu}}_2 - \hat{\boldsymbol{\mu}}_1). \end{aligned}$$

Simplifying the left-hand side

$$\begin{aligned}
\sum_{i=1}^N \mathbf{z}_i \mathbf{z}_i' &= \sum_{i=1}^N (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})' = \sum_{h=1}^2 \sum_{i \in h} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})' \\
&= \sum_{h=1}^2 \sum_{i \in h} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h + (\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}}))(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h + (\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}}))' \\
&= \sum_{h=1}^2 \sum_{i \in h} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h)' + \sum_{h=1}^2 \sum_{i \in h} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h)(\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})' \\
&\quad + \sum_{h=1}^2 \sum_{i \in h} (\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h)' + \sum_{h=1}^2 \sum_{i \in h} (\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})(\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})'.
\end{aligned}$$

Because $\sum_{i \in h} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h) = N_h \hat{\boldsymbol{\mu}}_h - N_h \hat{\boldsymbol{\mu}}_h = 0$, we get

$$\begin{aligned}
\sum_{i=1}^N \mathbf{z}_i \mathbf{z}_i' &= \sum_{h=1}^2 \sum_{i \in h} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_h)' + \sum_{h=1}^2 \sum_{i \in h} (\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})(\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})', \quad \text{and:} \\
\sum_{h=1}^2 \sum_{i \in h} (\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})(\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})' &= N_1 \mathbf{d}_1 \mathbf{d}_1' + N_2 \mathbf{d}_2 \mathbf{d}_2', \quad \text{where } \mathbf{d}_h = \hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}}.
\end{aligned}$$

Clearly $N_1 \mathbf{d}_1 + N_2 \mathbf{d}_2 = N_1 \hat{\boldsymbol{\mu}}_1 + N_2 \hat{\boldsymbol{\mu}}_2 - (N_1 + N_2) \bar{\mathbf{x}} = N \bar{\mathbf{x}} - N \bar{\mathbf{x}} = 0$, so that $\mathbf{d}_1 = -\frac{N_2}{N_1}(\hat{\boldsymbol{\mu}}_1 - \bar{\mathbf{x}})$.

Let $\mathbf{d} = \hat{\boldsymbol{\mu}}_2 - \hat{\boldsymbol{\mu}}_1 = \mathbf{d}_2 - \mathbf{d}_1$, so that

$$N_1 \mathbf{d}_1 + N_2 (\mathbf{d} + \mathbf{d}_1) = 0 \iff N \mathbf{d}_1 = -N_2 \mathbf{d} \iff \mathbf{d}_1 = -\frac{N_2}{N} \mathbf{d},$$

and similarly $\mathbf{d}_2 = \frac{N_1}{N} \mathbf{d}$. We then obtain

$$\sum_{h=1}^2 \sum_{i \in h} (\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})(\hat{\boldsymbol{\mu}}_h - \bar{\mathbf{x}})' = N_1 \frac{N_2^2}{N^2} \mathbf{d} \mathbf{d}' + N_2 \frac{N_1^2}{N^2} \mathbf{d} \mathbf{d}' = \frac{N_1 N_2}{N^2} (N_2 + N_1) \mathbf{d} \mathbf{d}' = \frac{N_1 N_2}{N} \mathbf{d} \mathbf{d}'.$$

Let $\hat{\Sigma}_B = \frac{N_1 N_2}{N^2} (\hat{\boldsymbol{\mu}}_2 - \hat{\boldsymbol{\mu}}_1)(\hat{\boldsymbol{\mu}}_2 - \hat{\boldsymbol{\mu}}_1)' = \frac{N_1 N_2}{N^2} \mathbf{d} \mathbf{d}'$, we then get that $\hat{\boldsymbol{\beta}}$ satisfies

$$\left[(N-2) \hat{\Sigma} + N \hat{\Sigma}_B \right] \hat{\boldsymbol{\beta}} = N(\hat{\boldsymbol{\mu}}_2 - \hat{\boldsymbol{\mu}}_1).$$

4 Equivalence between LS and LDA (two classes)

For any vector $\mathbf{v} \in \mathbb{R}^p$, $\hat{\Sigma}_B \mathbf{v} = \frac{N_1 N_2}{N^2} \mathbf{d}(\mathbf{d}' \mathbf{v})$, where $\mathbf{d}' \mathbf{v}$ is a scalar, so that $\hat{\Sigma}_B \mathbf{v}$ is a multiple of $\mathbf{d}$. Specifically, for $\mathbf{v} = \hat{\boldsymbol{\beta}}$, we get

$$\hat{\Sigma}_B \hat{\boldsymbol{\beta}} = \frac{N_1 N_2}{N^2} \mathbf{d}(\mathbf{d}' \hat{\boldsymbol{\beta}}) = \frac{N_1 N_2}{N^2} \mathbf{d}' \hat{\boldsymbol{\beta}} \mathbf{d} \equiv \alpha \mathbf{d},$$

where $\alpha = \frac{N_1 N_2}{N^2} \mathbf{d}' \hat{\boldsymbol{\beta}}$ is a scalar, so that $\hat{\Sigma}_B \hat{\boldsymbol{\beta}}$ is in the direction $\mathbf{d} = \hat{\mu}_2 - \hat{\mu}_1$. Substituting,

$$\begin{aligned} (N-2)\hat{\Sigma}\hat{\boldsymbol{\beta}} + N\hat{\Sigma}_B\hat{\boldsymbol{\beta}} &= N\mathbf{d} \iff (N-2)\hat{\Sigma}\hat{\boldsymbol{\beta}} + N\alpha\mathbf{d} = N\mathbf{d} \\ &\iff (N-2)\hat{\Sigma}\hat{\boldsymbol{\beta}} = N(1-\alpha)\mathbf{d} \\ &\implies \hat{\boldsymbol{\beta}} = \frac{N(1-\alpha)}{N-2}\hat{\Sigma}^{-1}\mathbf{d} \propto \hat{\Sigma}^{-1}(\hat{\mu}_2 - \hat{\mu}_1). \end{aligned}$$

Hence, the LS regression coefficient vector is identical to the LDA coefficient vector, up to a scalar multiple.

The intercept and decision boundary

Let $c = \frac{N(1-\alpha)}{N-2}$ so that $\hat{\boldsymbol{\beta}} = c\hat{\Sigma}^{-1}\mathbf{d}$. From the normal equation for the intercept we have

$$\hat{\beta}_0 = -\bar{\mathbf{x}}'\hat{\boldsymbol{\beta}} = -c\bar{\mathbf{x}}'\hat{\Sigma}^{-1}\mathbf{d}.$$

For LS, the predicted value is

$$\hat{f}(\mathbf{x}) = \hat{\beta}_0 + \mathbf{x}'\hat{\boldsymbol{\beta}} = -c\bar{\mathbf{x}}'\hat{\Sigma}^{-1}\mathbf{d} + c\mathbf{x}'\hat{\Sigma}^{-1}\mathbf{d} = c(\mathbf{x} - \bar{\mathbf{x}})'\hat{\Sigma}^{-1}\mathbf{d}.$$

Consider the LS rule where $\hat{f}(\mathbf{x}) > 0 \Rightarrow$ Class 2, and Class 1 otherwise. The decision boundary is then $(\mathbf{x} - \bar{\mathbf{x}})'\hat{\Sigma}^{-1}\mathbf{d} = 0$, i.e.

$$\mathbf{x}'\hat{\Sigma}^{-1}\mathbf{d} = \bar{\mathbf{x}}'\hat{\Sigma}^{-1}\mathbf{d},$$

while the LDA threshold is

$$\frac{1}{2}(\hat{\mu}_1 + \hat{\mu}_2)'\hat{\Sigma}^{-1}\mathbf{d} - \log(N_2/N_1) = \mathbf{x}'\hat{\Sigma}^{-1}\mathbf{d}.$$

Taking $\hat{\Sigma}$ p.d., then $\mathbf{d}'\hat{\Sigma}^{-1}\mathbf{d} > 0$; denote this quantity by $q \equiv \mathbf{d}'\hat{\Sigma}^{-1}\mathbf{d}$.

5 Comparing the LS and LDA boundaries

Let $a = \hat{\mu}_1'\hat{\Sigma}^{-1}\mathbf{d}$, then

$$\hat{\mu}_2'\hat{\Sigma}^{-1}\mathbf{d} = (\hat{\mu}_2 - \hat{\mu}_1 + \hat{\mu}_1)'\hat{\Sigma}^{-1}\mathbf{d} = (\hat{\mu}_1 + \mathbf{d})'\hat{\Sigma}^{-1}\mathbf{d} = a + q.$$

Now,

$$\begin{aligned} \underbrace{\bar{\mathbf{x}}'\hat{\Sigma}^{-1}\mathbf{d}}_{\text{(LS)}} &= \left(\frac{N_1}{N}\hat{\mu}_1 + \frac{N_2}{N}\hat{\mu}_2 \right)' \hat{\Sigma}^{-1}\mathbf{d} = \frac{N_1}{N}a + \frac{N_2}{N}(a+q) = a + \frac{N_2}{N}q, \\ \underbrace{\frac{1}{2}(a+a+q) - \log(N_2/N_1)}_{\text{(LDA)}} &= a + \frac{1}{2}q - \log(N_2/N_1). \end{aligned}$$

Taking the difference in boundary:

$$\begin{aligned} a + \frac{N_2}{N}q - a - \frac{1}{2}q + \log(N_2/N_1) &= \left(\frac{N_2}{N} - \frac{1}{2} \right) q + \log(N_2/N_1) \\ &= \frac{2N_2 - N}{2N}q + \log(N_2/N_1) = \frac{2N_2 - N_1 - N_2}{2N}q + \log(N_2/N_1) \\ &= \frac{N_2 - N_1}{2N}q + \log\left(\frac{N_2}{N_1}\right), \end{aligned}$$

hence the two rules are identical iff $N_1 = N_2$ (because of the intercept).

With more than 2 classes, LDA is not the same as linear regression, and it avoids the masking problems associated with that approach.